

# Applied Machine Learning Project Report

Project: Customer Churn Prediction in a Subscription-Based Service (Telco Churn)

Course: Applied Machine Learning

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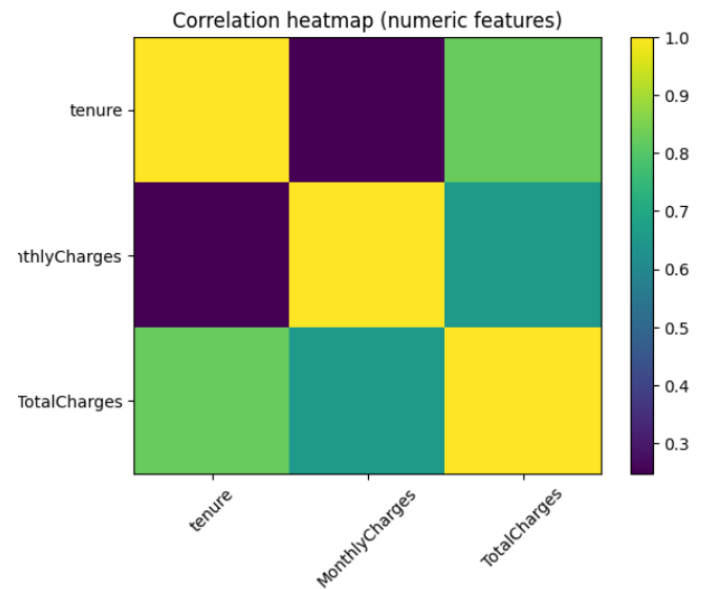
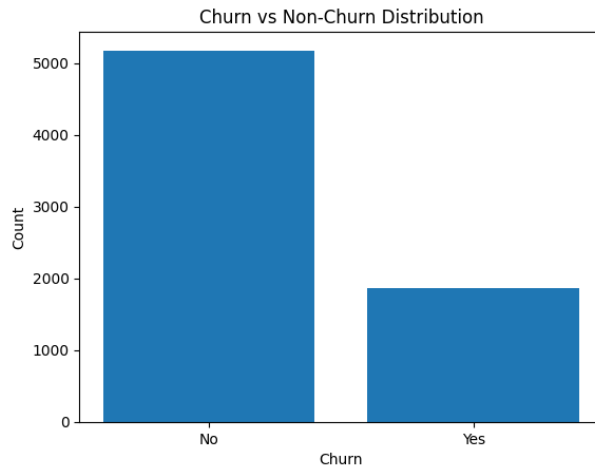
Dataset: Kaggle – Telco Customer Churn

## 1. Introduction

Customer churn is a critical issue for telecommunication companies, leading to significant revenue loss. Proactively identifying customers at risk of churning allows companies to implement targeted retention strategies. This project aims to build predictive models to forecast customer churn and to gain interpretable insights into the factors influencing churn.

## 2. Exploratory Data Analysis (EDA)

The Telco Customer Churn dataset contains 7,043 observations with 21 features, comprising both numerical and categorical variables. The target variable *Churn* is binary. During initial inspection, *TotalCharges* was found to be stored as an object due to empty string values. A total of 11 missing values were identified and corresponded to customers with zero tenure; these were imputed as 0 to reflect the absence of accumulated charges. Several categorical variables contained inconsistent labels such as “No internet service” and “No phone service,” which were consolidated to “No” to ensure consistent encoding. Distribution analysis revealed that *tenure* is right-skewed, with a large proportion of customers in their initial months and another concentration near the maximum contract duration, while *MonthlyCharges* exhibited slight bimodality, separating low-usage customers from those with higher service subscriptions. *TotalCharges* was highly right-skewed and strongly correlated with *tenure*. No significant outliers were detected using the IQR method. Correlation analysis showed a strong positive relationship between *tenure* and *TotalCharges* (0.83) and a moderate correlation between *MonthlyCharges* and *TotalCharges* (0.65). The dataset is moderately imbalanced, with 26.54% churned customers and 73.46% non-churned customers, justifying the use of F1-score as the primary evaluation metric. Further analysis indicated that customers with shorter tenure and lower total charges are significantly more likely to churn, while higher monthly charges—particularly among fiber optic users—are also associated with increased churn risk.



### 3. Data Preprocessing and Model Training (Shallow Models)

After data cleaning and category consolidation, the customerID column was removed as it carries no predictive value. The target variable *Churn* was encoded numerically (0 = No, 1 = Yes). The dataset was then split into training and test sets using stratified sampling to preserve the original churn distribution. Categorical features were transformed using one-hot encoding, while numerical features were standardized to ensure comparable feature scales. Several shallow learning models were trained and tuned using cross-validation. Logistic Regression was optimized using the liblinear solver with class-weight balancing and tuning of the regularization parameter  $C$ . The Random Forest classifier was trained with balanced class weights and tuned for key hyperparameters including the number of trees, tree depth, and minimum samples per split and leaf. Gradient Boosting was tuned by optimizing the number of estimators, learning rate, and tree depth. Finally, a soft Voting Ensemble was constructed by combining the best-performing Logistic Regression, Random Forest, and Gradient Boosting models, leveraging predicted class probabilities to improve overall predictive performance.

### 4. Neural Network (FFNN) and Autoencoder

For the neural network-based models, the data was preprocessed using one-hot encoding and standardization, followed by a split into training, validation, and test sets. Two neural network approaches were explored. First, a raw Feedforward Neural Network (FFNN) was trained on the full feature set, consisting of two dense layers with Batch Normalization, Dropout, and L2 regularization to reduce overfitting. Key hyperparameters such as activation function, learning rate, and optimizer were tuned experimentally. Second, an autoencoder was used to compress the original features into an 8-dimensional latent representation. These learned latent features were then used to train a Logistic Regression classifier for churn prediction, enabling a comparison between direct neural network learning and representation learning followed by a classical model.

## 5. Model Comparison and Analysis

| Model                                          | Accuracy | F1-score      |
|------------------------------------------------|----------|---------------|
| Voting Ensemble (soft)                         | 0.7807   | <b>0.6264</b> |
| Random Forest (tuned)                          | 0.7701   | 0.6241        |
| Logistic Regression (tuned)                    | 0.7395   | 0.6125        |
| Raw FFNN (23 features)                         | 0.7963   | 0.6008        |
| Autoencoder (8 features) + Logistic Regression | 0.7232   | 0.5963        |
| Gradient Boosting (tuned)                      | 0.8027   | 0.5813        |

Model performance comparison showed that the soft Voting Ensemble achieved the highest F1-score (0.6264), indicating the best balance between precision and recall for churn prediction in this imbalanced dataset. The tuned Random Forest closely followed with an F1-score of 0.6241, confirming the strength of ensemble tree-based methods. While Gradient Boosting and the raw FFNN achieved the highest accuracies (0.8027 and 0.7963 respectively), their lower F1-scores suggest a bias toward the majority non-churn class, making them less effective for identifying churners. The autoencoder-based approach performed the weakest, likely due to loss of discriminative information when compressing features from 23 to 8 dimensions.

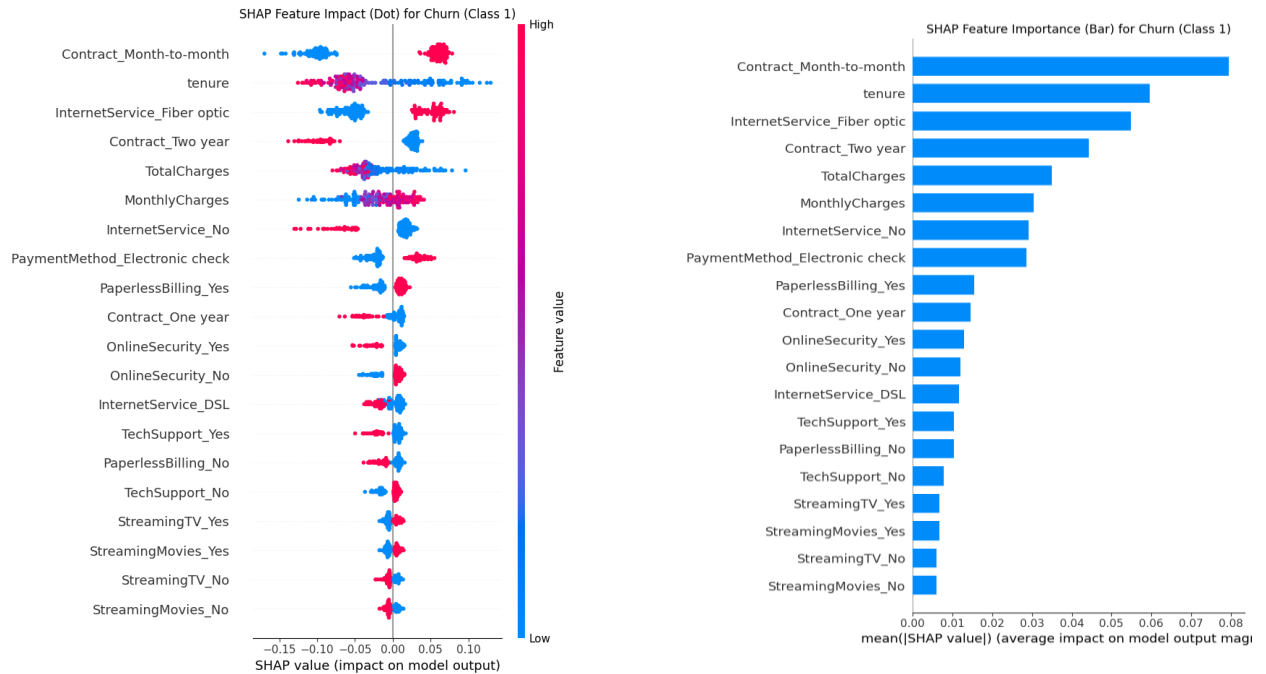
## 6. Misclassification Analysis (Voting Ensemble)

For the best-performing Voting Ensemble model, misclassification analysis on the test set revealed 194 false positives, where non-churn customers were incorrectly predicted to churn. These cases often involved customers with higher monthly charges, multiple subscribed services, and fiber optic internet, but also longer tenure or higher total charges, which may have introduced conflicting signals. Additionally, 115 false negatives were observed, representing churned customers incorrectly classified as non-churn; these customers typically had moderate tenure and monthly charges with less pronounced churn indicators. Analyzing these misclassifications helps identify patterns where the model struggles and provides direction for potential improvements through feature engineering, threshold adjustment, or refined model tuning.

## 7. SHAP Interpretability Analysis

SHAP values were used to interpret feature importance and impact for both the Random Forest and raw FFNN models. Both models consistently identified `Contract_Month-to-month`, `InternetService_Fiber optic`, `tenure`, `PaymentMethod_Electronic check`, `MonthlyCharges`, and `TotalCharges` as the most influential drivers of churn. `Month-to-month` contracts emerged as the strongest positive contributor to churn risk, while `tenure` and `TotalCharges` exhibited a clear non-linear inverse relationship, with churn risk sharply decreasing as customer duration and accumulated charges increased. Higher `MonthlyCharges`, fiber optic internet service, and electronic check payment methods were also associated with increased churn probability. Overall, both models showed strong agreement in feature importance and impact

direction, reinforcing the robustness of these insights. Minor differences were observed, with the FFNN appearing slightly more sensitive to tenure effects and producing smoother dependency patterns, while the Random Forest placed relatively greater emphasis on fiber optic service and exhibited more step-like behavior due to its tree-based structure.



## 8. Key Findings and Recommendations

The analysis revealed that customer churn is primarily driven by *Contract\_Month-to-month*, *InternetService\_Fiber optic*, *tenure*, *PaymentMethod\_Electronic check*, *MonthlyCharges*, and *TotalCharges*, with newer customers showing significantly higher churn risk. Customers on month-to-month contracts and fiber optic services are particularly prone to churn, highlighting areas requiring targeted intervention. Among all evaluated models, the soft Voting Ensemble achieved the most balanced predictive performance, delivering the highest F1-score and thus providing the most reliable identification of potential churners. From a business perspective, churn prevention efforts should focus on incentivizing long-term contracts, improving service quality and support for fiber optic customers, strengthening onboarding and early engagement for new users, reviewing issues associated with electronic check payments, and ensuring that customers with higher monthly charges perceive sufficient value. From a modeling standpoint, future improvements include refining ensemble strategies through stacking or weighted voting, addressing class imbalance using resampling or specialized loss functions, optimizing decision thresholds to align with business objectives, enhancing feature engineering, and continuously monitoring and retraining models to adapt to evolving customer behavior.